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Subsurface Engineering for Directional-Selective CO₂-to-Ethanol Electrocatalysis at Industrial-Level

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Abstract

The challenge in precisely controlling the adsorption configuration of oxygen-binding intermediates in the branching path following C-C coupling constrains the directed selectivity of electroreduction CO₂-to-ethanol. Here, we present a subsurface Co-doped CuS (Co-Sub-CuS) catalyst, which exhibits directed selectivity toward ethanol. We elucidate the role of subsurface doping in enhancing the oxophilicity of surface Cu sites, thereby facilitating the conversion of key intermediates (*CHCHO*) via the formation of surface-O bonds, guiding subsequent protonation towards ethanol. Moreover, the surface sulfur vacancies created by subsurface Co-doping help regulate the optimal distance between dual sites, facilitating asymmetric C-C coupling. Theoretical calculations combined with in-situ isotopic spectroscopy validate these views, and the branching pathway for converting *CHCO to *CHCHO* is captured. Consequently, in a membrane electrode assembly electrolyzer, the optimized Co-Sub-CuS achieves an ethanol Faradaic efficiency of 78.7% at a partial current density of 550.9 mA cm⁻², with stability over 305 h at industrial-level current density of 700 mA cm⁻². These findings provide a rational design for the development of directionally selective catalysts for CO₂ electroreduction.

Introduction

The efficient electrochemical reduction of carbon dioxide (CO₂ER) to high-value-added multicarbon (C₂) chemicals is a highly sought-after objective, positioned at the nexus of carbon neutrality and the clean energy economy¹⁻³. Among these C₂ products, ethanol (EtOH) is particularly promising due to its high energy density (26.8 MJ/kg) and ease of transport³⁻⁷. As is well known, Cu-based catalysts are among the most efficient and promising catalysts for electrochemically reducing CO₂ to ethanol⁸⁻¹¹. However, current documentation on the directional selectivity for EtOH on Cu-based electrocatalysts remains sparse, primarily owing to the notoriously sluggish C-C coupling kinetics and the unsettled bifurcation pathway involving a series of selective determining intermediates (SDI)¹²⁻¹⁵. In particular, the competing pathway for ethylene—another C₂ product—poses a significant challenge in controlling the bifurcation towards EtOH at the post-C-C coupling stage^{5,12,14}. Therefore, steering the reaction towards EtOH pathway after C-C coupling is crucial for achieving the desired directional selectivity for EtOH, alongside the ongoing focus on enhancing C-C coupling dynamics^{11,12,15}.

Experimental observations suggest that *CO intermediates undergo C-C coupling process, followed by protonation of the α -C to generate the shared initial intermediate *CHCO^{7,9,15,16}. The subsequent protonation pathway of *CHCO, which forms a variable SDI in a tautomeric form, may play a decisive role in dictating the selectivity between C₂ hydrocarbons (mainly C₂H₄) and C₂ oxygenates (mainly EtOH), as revealed by density functional theory (DFT) calculations^{10,15-17}. These variable SDI configurations can be broadly categorized based on the coordination of the

atoms with the Cu surface—either as $^*\text{CH}^*\text{COH}$ in the form of surface-C bonds¹⁷⁻¹⁹, or as $^*\text{CHCHO}^*$ in the form of surface-O bonds^{17,19}. Theoretically, the latter configuration could efficiently prevent the loss of O atom in subsequent steps, thereby favoring the formation of C₂ oxygenates^{6,8,17,19}. In contrast, the former configuration often occurs spontaneously due to its low free energy barrier, leading to dehydration during subsequent protonation steps and ultimately producing only C₂ hydrocarbons^{5,14,17-19}. In this regard, the binding strength of O atoms in the $^*\text{CHCHO}^*$ on the surface of Cu-based electrocatalysts could serve as a crucial descriptor for evaluating the bifurcation potential between C₂H₄ and EtOH formation. In current Cu-based catalysts system, however, the experimental identification and induction of $^*\text{CHCHO}^*$ from $^*\text{CHCO}$ have not been reported, primarily due to the uncontrollable binding of O atoms and the uncertainty surrounding the adsorption form of the SDI^{8-11,16,17}. Therefore, there is a pressing need to develop a rational and versatile catalyst design strategy to direct the branching SDI pathway toward EtOH, focusing on controlling the adsorption configuration of oxygen-binding intermediates ($^*\text{CHCHO}^*$).

To date, strategies aimed at modifying surface oxophilicity have provided alternative routes to influence the adsorption configuration of oxygen-binding intermediates through techniques such as surface engineering^{20,21}, doping effects²², alloying^{23,24}, and interface engineering²⁵. These approaches fundamentally rely on manipulating the electronic structure of active sites exposed at surface layer to optimize the coordination capacity of oxygen-binding intermediates. However, the electronic structure of active sites is largely determined by the local environment of the subsurface neighbors²⁶⁻²⁹. Thus, rational engineering of the subsurface in Cu-based catalysts offers a

promising approach to creating specific electronic states, thereby enabling precise regulation of surface oxophilicity and inducing the branching SDI, $^*\text{CHCHO}^*$.

In this work, we developed a facile and scalable strategy to synthesize subsurface Co-doped CuS catalysts (Co-Sub-CuS) featuring abundant sulfur vacancies (V_s) and surface-subsurface electron interactions, enabling the directional selectivity of CO_2 electroreduction to EtOH through the branching pathway from $^*\text{CHCO}$ to $^*\text{CHCHO}^*$ following promoted C-C coupling. Specifically, the intrinsically generated V_s and the specific electronic properties of Co-Sub-CuS were proven to regulate optimally-distanced dual sites and create asymmetric active sites, thereby triggering asymmetric OC-COH coupling due to the blended adsorption configuration. The electron transfer from Cu to Co, induced by subsurface Co doping, enhances oxophilicity at surface metal sites, promoting the formation and conversion of $^*\text{CHCHO}^*$ intermediates via surface-O bonds while suppressing the production of $^*\text{CH}^*\text{COH}$ intermediates. DFT predictions combined with in-situ isotopic spectroscopy validate these findings, capturing the selective determining intermediates $^*\text{CHCHO}^*$. Consequently, in a flow cell, the optimized Co-Sub-CuS catalysts exhibited a Faradaic efficiency (FE) of 87.6% and 79.4% for C_2 and EtOH products at a partial current density of 602.0 mA cm^{-2} , with a stability over 410 h. In a membrane electrode assembly (MEA) electrolyzer, the $\text{Co}_{0.050}\text{-Sub-CuS}$ catalysts achieved an FE of 78.7% for EtOH products, and a full-cell ethanol energy efficiency of 26.1% at -3.47 V. The concept of branching control over oxygen-binding intermediates, demonstrated using a generalized subsurfaced doped Cu-based catalyst model, marks a significant advancement in the directional CO_2 -to-EtOH conversion at industrial current densities. This approach offers valuable insights for both fundamental research and applications in

electroreduction CO₂ for fuel conversion, with broad and far-reaching implications.

Results

Preparation and Characterizations of Co-Sub-CuS Catalyst

Subsurface Co-doped CuS catalysts were synthesized via a one-step solvothermal method using [Cu(Tu)]Cl·1/2H₂O as the self-sacrificing template and Co³⁺ as the dopant. During the redox and vulcanization process, Co³⁺ was reduced to Co²⁺ and introduced into the subsurface of CuS (denoted as Co_x-Sub-CuS, where x is the atomic ratio of Co/(Co+Cu)). For comparison, Co-doped CuS surface catalysts (denoted as Co-Sur-CuS) were prepared through ion exchange using Co²⁺ as the dopant. Scanning electron microscopy (SEM), transmission electron microscopy (TEM), and X-ray diffraction (XRD) revealed that these catalysts exhibit similar nanosheet morphologies and CuS crystal structures (Supplementary Fig. 1a-d, Fig. 1a and Supplementary Figs. 2-4). The Co content in Co_x-Sub-CuS and Co-Sur-CuS catalysts was determined by inductively coupled plasma atomic emission spectroscopy (ICP-AES) and energy-dispersive X-ray (EDX) mapping (Supplementary Fig. 5, Supplementary Fig. 1e-h, and Supplementary Table 1). High-resolution TEM (HRTEM) measurements (Fig. 1b) showed that Co_{0.050}-Sub-CuS displays a lattice fringe of 3.24 Å (Fig. 1b1), indexed to the (100) planes of hexagonal CuS, which aligns with the corresponding fast Fourier transformation (FFT) (Fig. 1c and Supplementary Fig. 6). In contrast to the well-monocrystalline nature of pristine CuS and Co_{0.050}-Sur-CuS (Supplementary Fig. 6 and Fig. 7), Co_{0.050}-Sub-CuS exhibits numerous defects and disordered lattice fringes across the entire

nanocrystal surfaces (Fig. 1b2-b3 and Supplementary Fig. 8), along with additional diffraction spots in the FFT pattern, indicating a disruption of crystal symmetry (Fig. 1d and 1e). Enhanced signals at $g=2.003$ in electron paramagnetic resonance (EPR) spectroscopy (Supplementary Fig. 9) further confirmed a higher concentration of S-vacancies (V_s) in $\text{Co}_{0.050}\text{-Sub-CuS}$ compared to pristine CuS and $\text{Co}_{0.050}\text{-Sur-CuS}$. This defect-rich nature is likely a result of Co doping via redox reactions, which more effectively disrupts the original lattice compared to ion exchange doping.

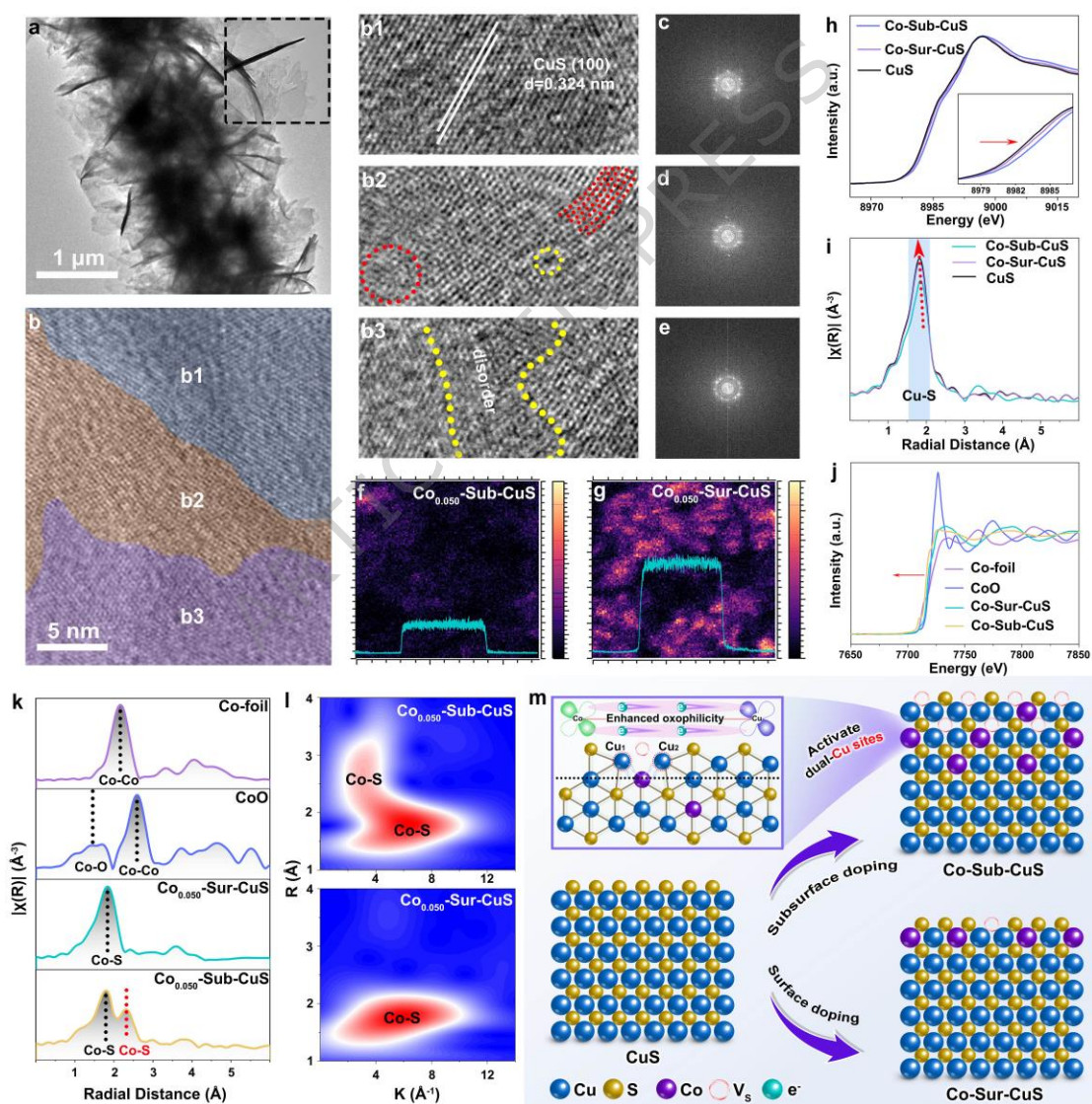


Fig. 1 | Structural characterizations. (a) TEM images of Co_{0.050}-Sub-CuS. (b) HRTEM images of Co_{0.050}-Sub-CuS. (b1-b3) enlargement of dash squares in (b), S-vacancies (V_S) (labeled by the yellow dotted circles) and line defect (marked by the dotted red line). (c-e) FFT patterns of respective region. Cobalt TOF-SIMS mapping images of Co_{0.050}-Sub-CuS (f) and Co_{0.050}-Sur-CuS (g), (insert, semiquantitative analysis of TOF-SIMS spectra of Co element of Co_{0.050}-Sub-CuS and Co_{0.050}-Sur-CuS). (h) Cu K-edge XANES spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS and CuS. (i) Fitting of k^2 -weighted EXAFS spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS and CuS. (j) Co K-edge XANES spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS and standards. (k) Fitting of k^2 -weighted EXAFS spectra of Co_{0.050}-Sur-CuS, Co_{0.050}-Sub-CuS and standards. (l) Corresponding wavelet transforms of k^2 -weighted EXAFS spectra at Co K-edge for Co_{0.050}-Sub-CuS and Co_{0.050}-Sur-CuS. (m) Scheme of activating and enhancing oxophilicity of surface Cu sites by subsurface Co doping, blue, yellow and purple color spheres in the model represent Cu, S and Co atoms, red-hollow and cyan-color spheres represent V_S and e⁻, respectively. Source data for Fig. 1h-l are provided as a Source Data file.

Subsurface Doping and the Optimization of Electronic Structure

To precisely determine the doping location of Co atoms, time-of-flight secondary-ion mass spectrometry (TOF-SIMS) was utilized, a highly sensitive method capable of detecting the top atomic layers of elements on the catalyst surface³⁰. Fig. 1f, 1g, and Supplementary Fig. 10 show more bright purple spots corresponding to Co elements in Co_{0.050}-Sur-CuS than in Co_{0.050}-Sub-CuS, which is consistent with semiquantitative analysis (Supplementary Fig. 11). This indicates that a greater proportion of Co atoms may exist in the subsurface of Co_{0.050}-Sub-CuS.

X-ray photoelectron spectroscopy (XPS) was employed to examine the valence states and electronic interactions among the elements in our catalysts (Supplementary Fig. 12). Supplementary Fig. 13a displays the Cu 2*p* core-level spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS, and CuS samples. In Co_{0.050}-Sub-CuS, the binding energies of 934.12 and 954.08 eV correspond to Cu²⁺ species, while the S 2*p* core-level spectra (Supplementary Fig. 13b) show typical S²⁻ species features at 161.90 and 163.19 eV^{31,32}. The Cu 2*p* peak of Co_{0.050}-Sub-CuS exhibits a

positive energy shift of 0.44 eV for Cu^{2+} species compared to CuS, whereas the S 2p peak shifts to lower binding energy by 0.38 eV. Similar XPS peak shifts were observed for other $\text{Co}_x\text{-Sub-CuS}$ samples (Supplementary Figs. 14-16), with the degree of shift increasing with a higher Co doping, peaking at $x=0.050$, which showed the maximum deviation value (Supplementary Fig. 17). In contrast, the XPS peak of the Co 2p in $\text{Co}_{0.050}\text{-Sub-CuS}$ exhibited the most pronounced shift toward the negative binding energy direction. This result indicates that $\text{Co}_{0.050}\text{-Sub-CuS}$ experienced the most significant electronic modulation within the $\text{Co}_x\text{-Sub-CuS}$ series³³.

The fine structure of Cu and Co species was further investigated by synchrotron X-ray absorption spectroscopy (XAS). The Cu valence states in $\text{Co}_{0.050}\text{-Sub-CuS}$, $\text{Co}_{0.050}\text{-Sur-CuS}$, and CuS were assessed via Cu K-edge X-ray absorption near-edge structure (XANES) spectra. Fig. 1h shows that both $\text{Co}_{0.050}\text{-Sub-CuS}$ and $\text{Co}_{0.050}\text{-Sur-CuS}$ exhibit absorption edges shifted to higher energy compared to pristine CuS, with $\text{Co}_{0.050}\text{-Sub-CuS}$ exhibiting a larger shift, indicating higher oxidation states of Cu atoms. The extended X-ray absorption fine structure (EXAFS) spectra at the Cu K-edge (Fig. 1i and Supplementary Fig. 18) reveal primary peaks at 1.81 Å, which are consistent across $\text{Co}_{0.050}\text{-Sub-CuS}$, $\text{Co}_{0.050}\text{-Sur-CuS}$, and pristine CuS, indicating the main structure of CuS is maintained. However, the peak intensity in $\text{Co}_{0.050}\text{-Sub-CuS}$ is weaker, suggesting a higher degree of coordination defects and surface disorder^{34,35}. The Co absorption edge in $\text{Co}_{0.050}\text{-Sub-CuS}$ is significantly shifted to a lower energy compared to $\text{Co}_{0.050}\text{-Sur-CuS}$ (Fig. 1j), indicating lower oxidation states of Co atoms. This shift is attributed to subsurface-doped Co inducing electron transfer from Cu to Co, aligning with the XPS results. The EXAFS spectra of the Co-edge (Fig. 1k and Supplementary Fig. 19) reveal a peak at 1.83 Å corresponding to Co-

S bonds in both Co_{0.050}-Sub-CuS and Co_{0.050}-Sur-CuS. However, a additional peak at 2.23 Å in Co_{0.050}-Sub-CuS suggests the presence of subsurface Co-S bonds. EXAFS wavelet transform analysis further confirms the existence of two types of Co-S bonds in Co_{0.050}-Sub-CuS (Fig. 1l). To sum up, as shown in Fig. 1m, subsurface doping of Co atoms generated surface defects that activated dual-Cu sites. Additionally, the electron transfer from Cu to Co sites induced by subsurface doping enhanced the oxophilicity of Cu sites, which was critical for the adsorption of key intermediates in the EtOH pathway during CO₂RR via surface-O bonds.

Catalytic Performance for CO₂ER

The catalytic performances were first evaluated in a three-electrode flow-cell using 1M KOH aqueous as the electrolyte (Supplementary Fig. 20). All voltages reported were converted to the reversible hydrogen electrode (RHE) scale with 85% iR-correction. Fig. 2a displays the linear sweep voltammetry (LSV) curves of the synthesized catalysts, measured in either CO₂- or Ar-saturated electrolytes. The cathodic current densities of these catalysts under CO₂-saturated conditions were largely higher than those under Ar-saturated conditions, highlighting the catalyst's preference for CO₂ER³⁶.

The Faradaic efficiencies (FEs) of various gaseous and liquid products (Supplementary Figs. 21-23) were measured within the potential range of -0.4 V to -1.0 V using gas chromatography (GC, Supplementary Fig. 24) and nuclear magnetic resonance (NMR, Supplementary Figs. 25-26). According to the FE of C₂ products for these catalysts under different applied potentials (Fig. 2b), EtOH was the exclusive C₂ oxygenate, while C₂H₄ and C₂H₆ were the only C₂ hydrocarbons

(mainly C_2H_4). As the applied potential increased, the FE_{C_2} of $Co_{0.050}$ -Sub-CuS exhibited a volcano-shaped trend, reaching a maximum value of approximately 90% at -0.8 V. Additionally, the FE_{EtOH} reached around 80% among all the products, demonstrating a high selectivity towards EtOH. $Co_{0.050}$ -Sub-CuS also exhibited the highest total current density and partial current density for both C_2 and EtOH across the applied potentials. Notably, the partial current density of EtOH on $Co_{0.050}$ -Sub-CuS reached 602.0 mA cm^{-2} at -0.8 V, significantly surpassing that of $Co_{0.050}$ -Sur-CuS (140.8 mA cm^{-2}) or CuS (49.9 mA cm^{-2}) (Fig. 2b and Supplementary Figs. 27-28), exceeding the current target for practical applications ($> 300 \text{ mA cm}^{-2}$)^{37,38}.

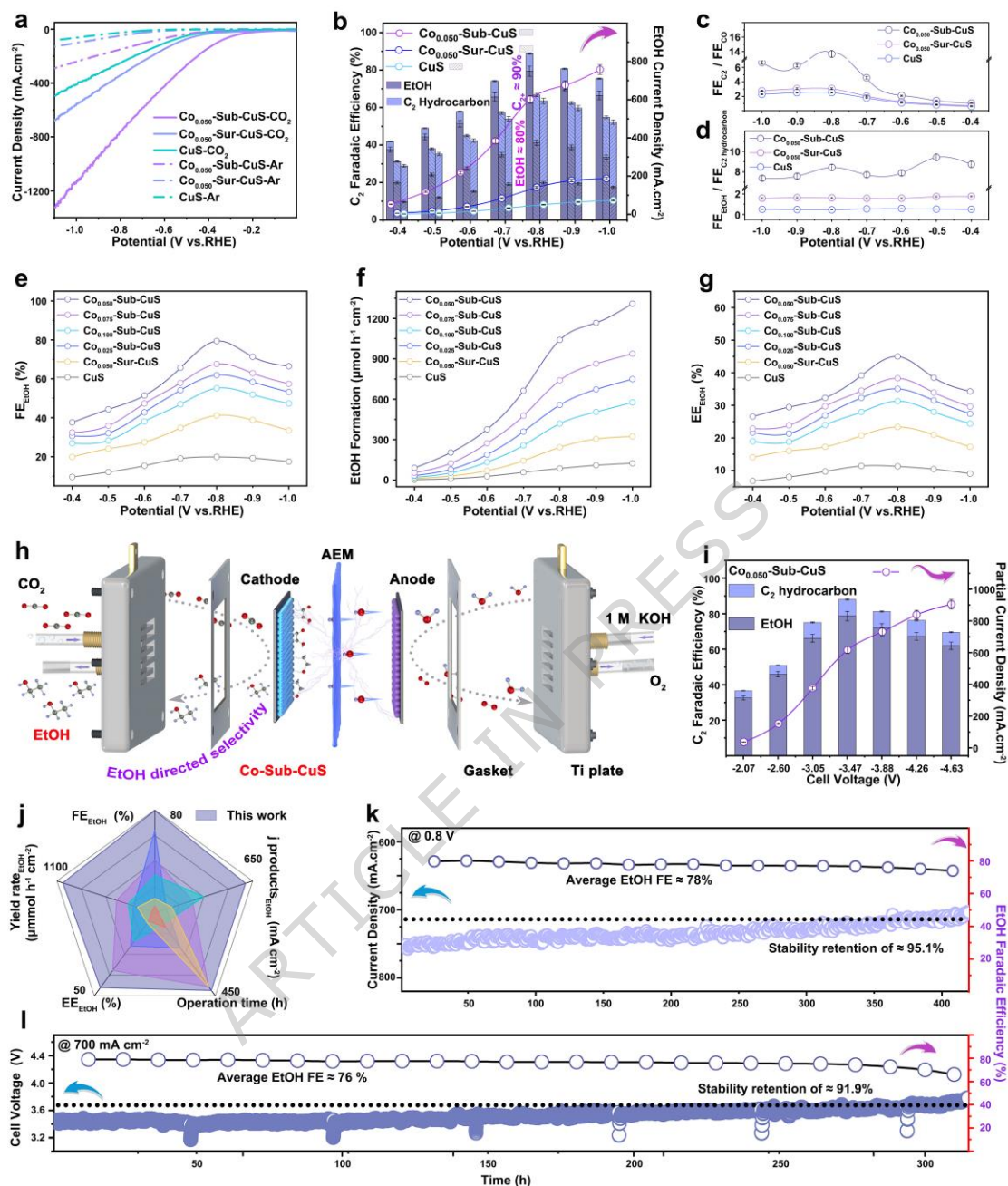


Fig. 2 | CO₂ER performance for Co_x-Sub-CuS, Co_{0.050}-Sur-CuS and CuS. (a) LSV curves. (b) Potential-dependent C₂ FE and EtOH current density. (c) Ratio of FE_{C2} to FE_{CO} and (d) Ratio of FE_{EtOH} to FE_{C2H4} under selected potential. FE (e), yield rate (f) and EE (g) for EtOH production over Co_x-Sub-CuS and Co_{0.050}-Sur-CuS. (h) Schematic diagram of the MEA electrolyzer. (i) C₂ FE and partial current density at selected cell voltage under MEA measurement. (j) Comparison of the performance metrics of the Co_{0.050}-Sub-CuS catalyst with recently reported catalysts (details in Table S8). (k) CO₂ER stability of the Co_{0.050}-Sub-CuS catalyst in a three-electrode flow cell at a

constant potential of 0.8 V vs. RHE. (l) EtOH FE and cell voltage during CO₂ER at a constant current of 700 mA cm⁻² in an MEA electrolyzer. All voltages were converted to the reversible hydrogen electrode (RHE) scale with 85% iR-correction. Error bars represent the standard deviation from at least three independent measurements. Source data for Fig. 2a-g, Fig. 2i and Fig. 2k, l are provided as a Source Data file.

Since *CO is considered a key intermediate for generating $\geq 2e^-$ products³⁹, the FE_{C2}/FE_{CO} ratio of these catalysts was calculated to assess the electrochemical conversion efficiency of *CO-to-C₂ (Fig. 2c). Co_{0.050}-Sub-CuS exhibiting a high FE_{C2}/FE_{CO} ratio at -0.8 V, nearly 6 and 9 times higher than that of Co_{0.050}-Sur-CuS and CuS, respectively, displayed the high *CO-to-C₂ conversion efficiency. The FE_{EtOH}/FE_{C₂H₄} ratio of Co_{0.050}-Sub-CuS was also the highest among these catalysts (Fig. 2d), demonstrating the high selectivity for EtOH. Co_{0.050}-Sub-CuS not only accelerated the C-C coupling process of *CO, enhancing the overall selectivity of C₂ products, but also promoted the directional preference for EtOH pathways in CO₂ER. In contrast, Co_{0.050}-Sur-CuS exhibited a higher FE_{EtOH}/FE_{C₂H₄} ratio than CuS, yet both catalysts demonstrated low *CO-to-C₂ conversion efficiencies. This suggests that Co doping promotes the selection of the EtOH path following C-C coupling, with surface V_s caused by subsurface Co doping likely playing a key role in improving the overall selectivity for C₂ product.

The CO₂ER performance of various Co_x-Sub-CuS catalysts with different Co doping levels was assessed (Supplementary Figs. 29-34, summarized in Supplementary Tables 2-7). It is shown that Co_{0.050}-Sub-CuS exhibited the highest FE_{EtOH} (Fig. 2e), EtOH formation rate (Fig. 2f), energy efficiency of EtOH (EE_{EtOH}, Fig. 2g), and EtOH partial current density (Supplementary Fig. 35) across different applied potentials, with a distinct inhibitory effect on FE_{C2} hydrocarbon compared to bare CuS (Supplementary Fig. 36), the heatmaps displayed more clearly (Supplementary Fig.

37). Both FE_{EtOH} and FE_{C_2} exhibited a volcanic trend with increasing Co doping, while FE_{CO} and FE_{C_2} hydrocarbon showed an inverse trend (Supplementary Figs. 38-40). For Cu-based catalysts, the *CO coverage on the Cu surface facilitates C-C coupling, affecting both EtOH and C_2H_4 formation. However, only FE_{EtOH} correlated with FE_{CO} on Co_x -Sub-CuS, indicating that enhanced EtOH selectivity not only depends on C-C coupling, but also on additional factors that direct the subsequent CO_2ER pathway toward EtOH.

Long-term stability remains a critical challenge for CO_2ER , especially at high current densities ($> 300 \text{ mA cm}^{-2}$). Fig. 2k shows that the $Co_{0.050}$ -Sub-CuS catalyst maintained stable electrolysis for over 410 h with a constant current density of $760\text{-}730 \text{ mA cm}^{-2}$ in a flow cell at -0.8 V , achieving a FE_{EtOH} of approximately 78%. This stability is attributed to the catalyst's robust structural integrity during CO_2ER (Supplementary Figs. 41-42). $Co_{0.050}$ -Sub-CuS also retained high hydrophobicity throughout the CO_2ER process, which is essential for maintaining efficient three-phase interfaces (Supplementary Figs. 43-44)⁴⁰. The performance and stability of $Co_{0.050}$ -Sub-CuS for EtOH production in CO_2ER meet commercial requirements and are competitive with the previously reported results (Fig. 2j and Supplementary Table 8).

To further explore the intrinsic electrocatalytic activity, the electrochemically active surface area (ECSA) was evaluated by electrochemical double-layer capacitance (C_{dl})²⁰. $Co_{0.050}$ -Sub-CuS exhibited the largest ECSA, which is favorable for the formation of abundant active sites for CO_2ER (Supplementary Figs. 45-46). The normalized ECSA current density of $Co_{0.050}$ -Sub-CuS was also the highest, proving its high intrinsic electrocatalytic activity (Supplementary Fig. 47). Additionally, the lowest Tafel slope and electrochemical impedance spectroscopy (EIS) results

demonstrated the fastest CO₂ER kinetics and charge transfer rate on Co_{0.050}-Sub-CuS^{22,41}, respectively (Supplementary Figs. 48-49).

The CO₂ER activity of Co_{0.050}-Sub-CuS was tested in a MEA electrolyzer (Fig. 2h). A strong linear relationship was observed between the applied cell voltage and current density (Supplementary Fig. 50). The FEs of all products tested in the flow-cell were well reproduced in the MEA electrolyzer (Fig. 2i and Supplementary Figs. 51-52, summarized in Supplementary Table 9). At a cell voltage of -3.47 V, Co_{0.050}-Sub-CuS catalysts achieved a FE_{EtOH} of 78.7%, a partial current density of $\approx 550.9 \text{ mA cm}^{-2}$ for EtOH, and a full-cell energy efficiency (EE_{full-cell}) of $\approx 26.1\%$ for EtOH (Supplementary Fig. 53), compared with previously reported electrocatalysts for selective ethanol production in an MEA electrolyzer, exhibited competitive electrocatalytic performance (Supplementary Table 10). The FE_{C2}/FE_{CO} ratio at different cell voltages was also evaluated (Supplementary Fig. 54), with a ratio of ≈ 13 at -3.47 V, indicating the high C-C coupling orientation capability. The CO₂ER stability of Co_{0.050}-Sub-CuS catalysts in MEA electrolyzers was systematically evaluated (Fig. 2l), showing well stability for 305 h at a total current density of 700 mA cm⁻², with a stable cell voltage ranging from -3.45 V to -3.72 V, and an average FE_{EtOH} of $\approx 76\%$. The microstructure of Co_{0.050}-Sub-CuS remained intact after CO₂ER (Supplementary Figs. 55-56).

Deciphering Ethanol Reaction Pathways

EtOH and C₂H₄ follow similar CO₂ER pathways, both undergoing protonation to the shared intermediate ^{*}CHCO, where the bonding configuration of its protonated form ^{*}CHCHO plays a

pivotal role in determining the selectivity between EtOH and C_2H_4 ^{11,16,17}. The selectivity of $Co_{0.050}$ -Sub-CuS for EtOH in CO_2 ER suggests that subsurface Co doping may alter the bonding mode of *CHCHO , thereby steering the reaction towards the selective production of EtOH. In-situ Fourier-transform infrared (FTIR) spectroscopy was employed to dynamically track the evolution of reaction intermediates, complemented by DFT calculations to explore potential bonding modes of intermediates in the CO_2 -to-EtOH transformation pathway.

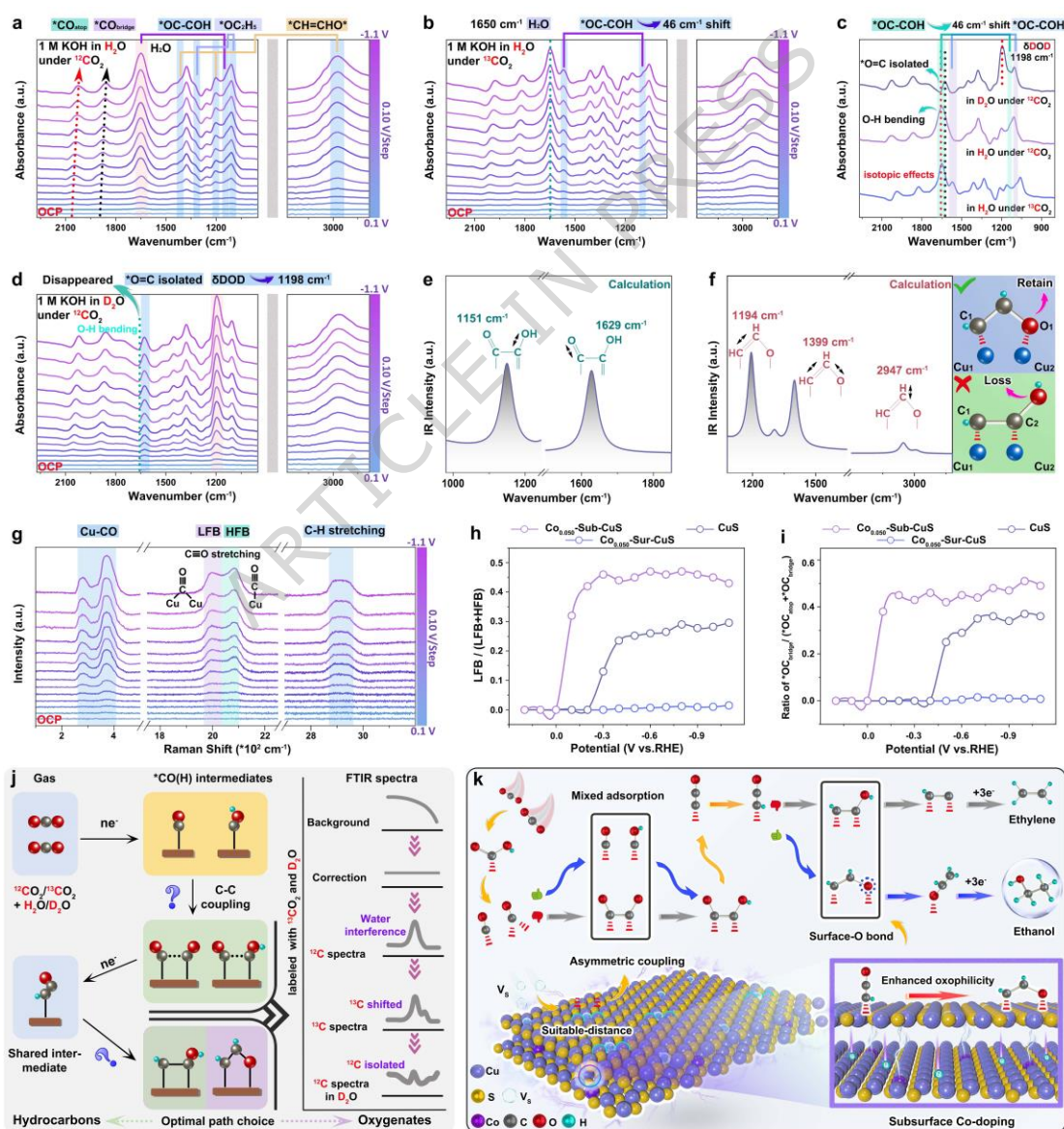


Fig. 3 | In-situ characterization and mechanism exploration of the CO₂ER to EtOH. In-situ FTIR spectra of Co_{0.050}-Sub-CuS in (a) ¹²CO₂-saturated 1 M KOH with H₂O, (b) ¹³CO₂-saturated 1 M KOH with H₂O. (c) Comparison of the in-situ FTIR spectra on Co_{0.050}-Sub-CuS in ¹²CO₂-saturated 1 M KOH with H₂O, ¹³CO₂-saturated 1 M KOH with H₂O and ¹²CO₂-saturated 1 M KOH with D₂O. In-situ FTIR spectra of Co_{0.050}-Sub-CuS in (d) ¹²CO₂-saturated 1 M KOH with D₂O as electrolyte from 0.1 V vs. RHE to -1.1 V vs. RHE. (e) The calculated infrared-active vibrational frequency of *OCCOH on Co_{0.050}-Sub-CuS. (f) The calculated infrared-active vibrational frequency of *CHCHO* on Co_{0.050}-Sub-CuS, the right is the different adsorption modes of two key intermediates *CHCHO* and *CHC*OH at dual-Cu sites. (g) In-situ Raman peaks on Co_{0.050}-Sub-CuS in 1 M KOH from 0.1 V vs. RHE to -1.1 V vs. RHE during CO₂ER. (h) Comparison of the LFB/(HFB+LFB) ratio in in-situ Raman spectra for different catalysts at different applied potential. (i) Comparison of the *OC_{bridge}/(^{*}OC_{bridge}+^{*}OC_{atop}) ratio in in-situ FTIR spectra for different catalysts at different applied potential. (j) Schematic illustration of the underexplored asymmetric coupling process and ethanol path directed selectivity problem in CO₂ electroreduction, and flow diagram illustrating the working principles of isotopic tracking of surface-adsorbed intermediates based on ¹²C/¹³C and H/D isotopes in tandem with in-situ FTIR spectra. (k) Schematic for boosted EtOH generation over Co_{0.050}-Sub-CuS. Blue, silver, pink, red, gray, and green-color spheres in the model represent Cu, S, Co, O, C, and H atoms, respectively. Source data for Fig. 3a-i are provided as a Source Data file.

Fig. 3a shows the in-situ FTIR spectra of Co_{0.050}-Sub-CuS in a ¹²CO₂-saturated 1 M KOH with H₂O under potential-dependent conditions. Peaks at 2054 and 1896 cm⁻¹ for Co_{0.050}-Sub-CuS and Co_{0.050}-Sur-CuS (Supplementary Fig. 57) correspond to atop-adsorbed *CO (*CO_{atop}) and bridge-adsorbed *CO (*CO_{bridge}) on Cu sites, respectively^{38,42-45}. In contrast, CuS mainly shows the *CO_{atop} peak, with minimal evidence of *CO_{bridge} in the same potential region (Supplementary Fig. 58). Previous studies indicate that protonation of *CO_{bridge} on the Cu surface is more energetically favorable than protonation of *CO_{atop}^{38,42}. The simultaneous presence of *CO and *COH (or *CHO) following *CO_{bridge} protonation can trigger asymmetric C-C coupling on Co_{0.050}-Sub-CuS. The appearance of *CHO intermediates at 1259 cm⁻¹ in the in-situ FTIR spectra of Co_{0.050}-Sub-CuS further supports this mechanism⁴².

In-situ FTIR spectra of $^*\text{OCCOH}$ intermediates involved in C-C asymmetric coupling were verified using hydroxypropanone, a compound with a hydroxyl group adjacent to the carbonyl group (OCCOH). The band detected at 1152 cm^{-1} on $\text{Co}_{0.050}\text{-Sub-CuS}$ during CO_2ER at -0.8 V is consistent with the C-OH of hydroxypropanone (Supplementary Fig. 59 and Fig. 3a). However, H_2O exhibits a broad O-H bending band in the range $1750\text{-}1550\text{ cm}^{-1}$ ^{46,47}, which may overlap the C=O vibration signal at 1615 cm^{-1} . To further confirm this, a spectrum recorded using isotopically labelled ^{13}CO and replacing the solvent with D_2O (Fig. 3b and 3j). Pre-labeled C-reaction intermediates showed a shift of approximately $43\text{-}47\text{ cm}^{-1}$ compared with ^{12}C , confirming that these bands originate from C-containing species of CO_2ER ⁴⁸. More importantly, the C=O stretching of $^*\text{OCCOH}$ in the ^{13}CO atmosphere shifted to 1575 cm^{-1} , separating it from the O-H bending band in H_2O compared to ^{12}C spectrum (Fig. 3c and Supplementary Figure 60a). The spectra recorded in a ^{12}CO atmosphere with D_2O supported these views (Fig. 3d and Supplementary Figure 60b, see details in Supplementary Note 1). Theoretical calculations of the FTIR spectra of $^*\text{OCCOH}$ intermediates adsorbed on the surface of $\text{Co}_{0.050}\text{-Sub-CuS}$ showed a C=O stretching at 1151 cm^{-1} , and a C-OH stretching at 1629 cm^{-1} , which correspond to the experimental values of 1621 cm^{-1} and 1132 cm^{-1} , respectively (Fig. 3e and Supplementary Fig. 61). A peak at 1540 cm^{-1} , indicative of $^*\text{OCCO}$ species, was detected on CuS and $\text{Co}_{0.050}\text{-Sur-CuS}$, a key intermediate in traditional C-C symmetric coupling using surface-adsorbed $^*\text{CO}$ ^{42,49,50}.

Stronger bands at 1335 cm^{-1} and 1117 cm^{-1} were detected on $\text{Co}_{0.050}\text{-Sub-CuS}$, which are attributed to $^*\text{OC}_2\text{H}_5$ ^{38,49,51-53}, a key intermediate in EtOH production (Fig. 3a). In contrast, only weak $^*\text{OC}_2\text{H}_5$ signals were detected, or none at all, on $\text{Co}_{0.050}\text{-Sur-CuS}$ and CuS , respectively

(Supplementary Figs. 57-58). This observation highlights the directional selectivity of the Co_{0.050}-Sub-CuS catalyst for EtOH. The selectivity between EtOH and C₂H₄ is primarily governed by the bonding mode of the protonation intermediate ^{*}CHCHO form ^{*}CHCO at Cu sites. Co_{0.050}-Sub-CuS's pronounced selectivity for EtOH and the electron transfer mechanism induced by Co doping suggest that Co doping optimizes the electron configuration of Cu sites and changes the bonding form of ^{*}CHCHO. Electron transfer from Cu to Co sites decreases the coordination number of Cu sites, thereby enhancing the coordination capacity of Cu and O sites and promoting the protonation of ^{*}CHCO to ^{*}CHCHO^{*} intermediates in the form of surface-O bonds (the right schematic illustration of Fig. 3f). And the subsequent protonation steps are carried out via adsorbed O atoms. This effectively prevents the protons in the electrolyte from attacking the O atoms, thereby inhibiting dehydration and the loss of oxygen atom, which facilitates the production of EtOH in subsequent reactions. Theoretical calculations of the FTIR spectra of ^{*}CHCHO^{*} intermediates adsorbed on the surface of Co_{0.050}-Sub-CuS confirmed the protonation of ^{*}CHCO to ^{*}CHCHO^{*} (Fig. 3f and Supplementary Fig. 62). The calculated spectrum showed the C-H stretching in different directions at 1194 cm⁻¹, 1399 cm⁻¹ and 2497 cm⁻¹, which corresponded to the measured results at 1206 cm⁻¹, 1409 cm⁻¹ and 2979 cm⁻¹, respectively (see details in Supplementary Note 2). To further verify the key protonation process of ^{*}CHCO from ^{*}CHCHO^{*}, methoxyethene with the group of -CH=CH-O- was used. Band detected at 1190 cm⁻¹, 1417 cm⁻¹ and 2998 cm⁻¹ were corresponded to the experimental results (Supplementary Fig. 63).

Based on the above analysis, a mechanism for the directional selection of EtOH production can be proposed. Subsurface Co doping induces a mixed adsorption configuration of ^{*}CO_{atop} and

$^*\text{CO}_{\text{bridge}}$, resulting in asymmetric OC-COH coupling, which is further supported by in-situ Raman spectroscopy (Fig. 3g-i and Supplementary Figs. 64-67, see details in Supplementary Note 3). Subsequently, the shared intermediate $^*\text{CHCO}$ transformed into $^*\text{CHCHO}^*$ via surface-O bonds by hydrogenated carbon atoms, followed by the next protonation step through the adsorption of O atoms. The complete reaction pathway from CO_2 to EtOH on $\text{Co}_{0.050}\text{-Sub-CuS}$ is as follows: $2\text{CO}_2 \rightarrow 2^*\text{COOH} \rightarrow ^*\text{CO}_{\text{bridge}} + ^*\text{CO}_{\text{atop}} \rightarrow ^*\text{COH} + ^*\text{CO}_{\text{atop}} \rightarrow ^*\text{OCCOH} \rightarrow ^*\text{CCO} \rightarrow ^*\text{CHCO} \rightarrow ^*\text{CHCHO}^* \rightarrow ^*\text{OCHCH}_2 \rightarrow ^*\text{OCHCH}_3 \rightarrow ^*\text{OCH}_2\text{CH}_3 \rightarrow \text{C}_2\text{H}_5\text{OH}$ (Fig. 3k). The design of subsurface Co-doped CuS selectively directs EtOH production by promoting asymmetric C-C coupling and controlling the branching pathway.

DFT Calculations

To investigate the role of subsurface cobalt in controlling the selectivity of C_2 products, DFT calculations were performed. The analysis of CO_2ER selectivity begins with examining the co-adsorbed state of $^*\text{CO}$ and $^*\text{COH}$, which is a critical initial step in the competing reaction pathways^{54,55}. As shown in Fig. 4a, these pathways diverge towards the production of either ethylene (Path 1) or EtOH (Path 2). *Ab initio* molecular dynamics (AIMD) were used to compare the structural characteristics of CuS-based catalysts (Supplementary Fig. 68), one with only sulfur vacancies (Vs-CuS) and another with sulfur vacancies plus cobalt atoms doped into the sub-layer (Co-Sub-CuS). The radial distribution function (gI) shows a first peak within 2.55 Å at ambient temperature (Supplementary Fig. 69). Upon adsorption of CO_2ER intermediates onto the catalyst, the Cu-Cu distances change. Specifically, when $^*\text{CO}$ and $^*\text{COH}$ are co-adsorbed on the Vs-CuS

catalyst, the Cu-Cu distance increases from 2.67 Å to 3.52 Å after the coupling reaction, yielding OCCOH* (Supplementary Fig. 70). In contrast, on Co-Sub-CuS, this distance changes from 2.53 Å to 3.16 Å under the same conditions (Fig. 4b). The energy profiles suggest that asymmetric C-C coupling is thermodynamically favorable on both Vs-CuS and Co-Sub-CuS catalysts (Fig. 4c), confirming the feasibility of asymmetric coupling reactions on these catalysts.

After asymmetric C-C coupling, the determination of the C₂ product depends on the subsequent reduction path of the *CHCO intermediate: Path 1 leads to ethylene, while Path 2 results in EtOH. To understand the mechanism underlying this selectivity, energy profiles for the critical steps in both reaction pathways were calculated (Fig. 4d). On Vs-CuS, the *CHCO intermediate undergoes two spontaneous steps for both paths (blue line). In contrast, on Co-Sub-CuS, Path 1 shows an initial energy increase (0.64 eV) before decreasing, while Path 2 consistently decreases (red line). This indicates that Co-Sub-CuS has a pronounced preference for Path 2, resulting in higher selectivity towards EtOH over ethylene. The difference in reactivity is mainly due to interactions between the catalysts and the *CHC*OH and *CHCHO* intermediates. To explore the bonding characteristics at the interface, the projected crystal orbital Hamilton population (-pCOHP) analysis was conducted. On Vs-CuS, *CHC*OH is adsorbed at the Cu₁ and Cu₂ sites with the anti-bonding orbitals below the Fermi level (Fig. 4e), suggesting that the *CHC*OH intermediate is stabilized by Vs-CuS. In contrast, on Co-Sub-CuS, the C-Cu₁ bond shows anti-bonding states crossing the Fermi level, indicating electronic instability of the *CHC*OH adsorbed on Co-Sub-CuS. For the *CHCHO* intermediate, anti-bonding orbitals appear at the Fermi level in both the C-Cu and O-Cu bonds at the copper sites of Vs-CuS and Co-Sub-

CuS catalysts (Supplementary Fig. 71), indicating an unstable electronic structure for the $^*\text{CHCHO}^*$ intermediate. Given these observations, the $^*\text{CHC}^*\text{OH}$ and $^*\text{CHCHO}^*$ intermediates might need to undergo backward oxidation or forward reduction to achieve a more stable electronic state. Therefore, various theoretical adsorption configurations for $^*\text{CHCO}$, $^*\text{CHC}^*\text{OH}$ and $^*\text{CHCHO}^*$ on Vs-CuS (Supplementary Figs. 72-74) and Co-Sub-CuS (Supplementary Figs. 75-77) were explored to gain a comprehensive understanding of the adsorption states and reaction selectivity.

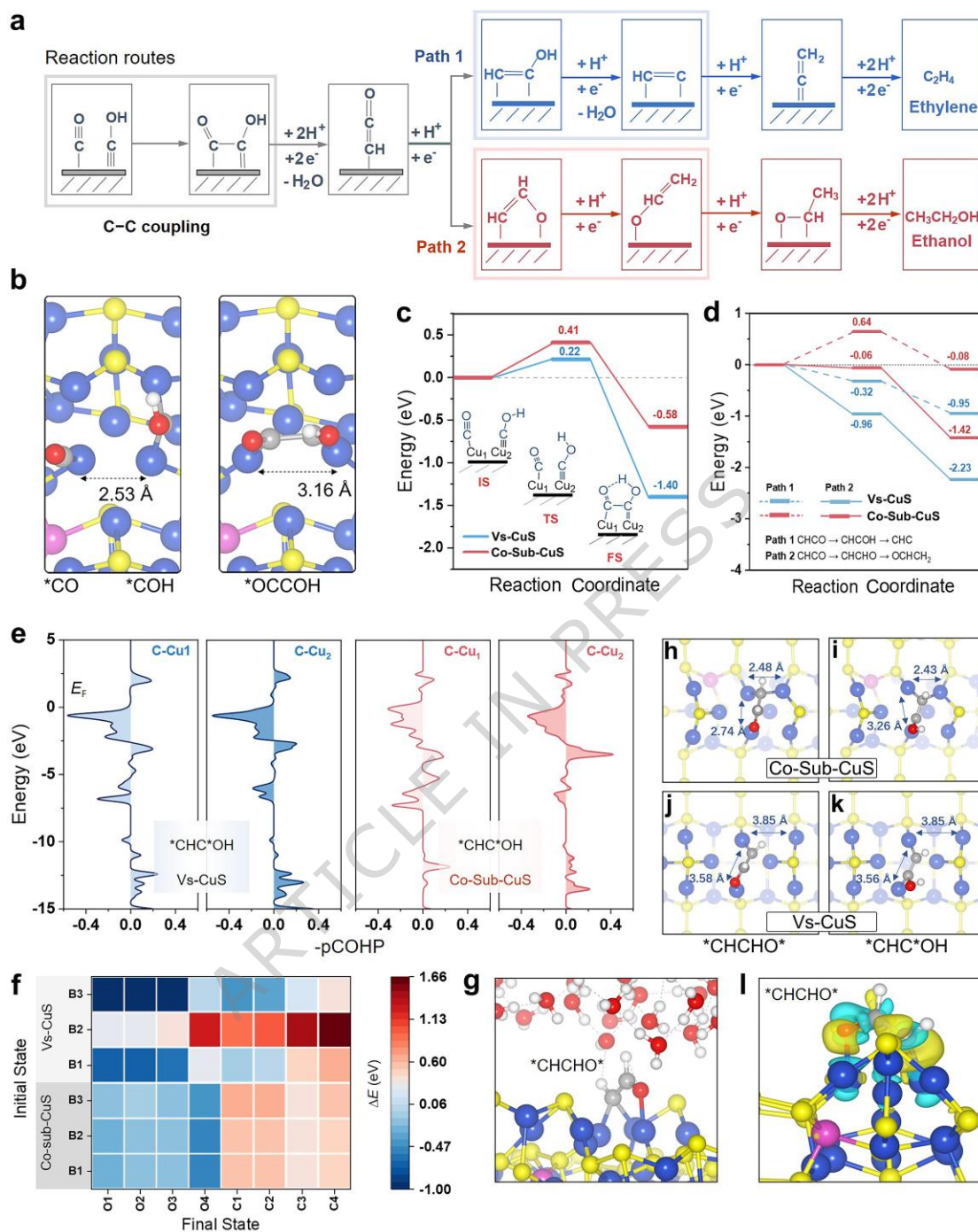


Fig. 4 | Mechanistic study. (a) Reaction routes via C-C coupling (grey) towards C_2 products. Path 1 for the generation of ethylene (blue), and Path 2 for EtOH (red). (b) On the Co-Sub-CuS catalyst, *CO and *COH co-adsorb on Cu sites and asymmetrically couple to form $OCCOH^*$, with dashed arrows denoting Cu-Cu distances. (c) Energy profiles for the $^*CO + ^*COH \rightarrow OCCOH^*$ on Vs-CuS and Co-Sub-CuS. Initial state (IS), transition state (TS) and final state (FS) are depicted in the

insets. (d) Energy profiles for the reduction of $^*\text{CHCO}$. (e) -pCOHP curves of the C-Cu₁ and C-Cu₂ bonds. (f) Heat map depicting reaction energy profiles for conversion of $^*\text{CHCO}$. B₁-B₃ refer to three adsorption configurations of $^*\text{CHCO}$; C₁-C₄ denote four configurations of $^*\text{CHC}^*\text{OH}$ along Path 1 and O₁-O₄ represent $^*\text{CHCHO}^*$ along Path 2. See Supplementary Information for structure details. (g) $^*\text{CHCHO}^*$ on Co-Sub-CuS with one layer of water molecules. Hydrogen bonds are shown in dashed lines, with arrows illustrating the Cu-Cu distance in (h) $^*\text{CHCHO}^*$ and (i) $^*\text{CHC}^*\text{OH}$. Configurations of (j) $^*\text{CHCHO}^*$ and (k) $^*\text{CHC}^*\text{OH}$ on Vs-CuS. (l) Differential charge density plots of $^*\text{CHCHO}^*$ on Co-Sub-CuS. Isosurfaces were generated at $\pm 0.002 \text{ e}^-/\text{bohr}^3$, with blue and yellow areas representing electron loss and gain, respectively. Source data for Fig. 4c-e are provided as a Source Data file.

Energy profiles for the transformation of $^*\text{CHCO}$ across different adsorption configurations were calculated (Fig. 4f). For the Vs-CuS catalyst, reaction energy changes exhibit a wide range from negative to positive (upper panel). In the C₁₋₄ region associated with Path 1, there is a coexistence of positive and negative energy variations, with a similar state in the O₁₋₄ region corresponding to Path 2. Therefore, on the Vs-CuS surface, whether Path 2 holds a selective advantage over Path 1 depends on the specific adsorption configuration. In contrast, with the Co-Sub-CuS catalyst, the competing reaction pathways show two distinct modes (lower panel). Specifically, reaction energies in the C₁₋₄ region are characterized by a positive distribution, whereas the O₁₋₄ region exhibits a negative distribution. Adsorbed intermediates B₁ and B₂, originating from distinct configurations, converge to the same optimized structure. Therefore, on the Co-Sub-CuS surface, Path 2 consistently exhibits a thermodynamic preference over Path 1, regardless of the adsorption configurations. To verify the preference of Path 2 for Co-Sub-CuS, water molecules were introduced into the adsorption configurations (Supplementary Fig. 78), allowing for calculations under conditions more closely resembling the electrode-solution interface. Hydrogen bonding interactions were observed between interfacial water and the

adsorbed intermediates $^*\text{CHCHO}^*$ (Fig. 4g) and $^*\text{CHC}^*\text{OH}$ (Supplementary Fig. 79). For the Co-Sub-CuS catalyst, three Cu sites participate in the adsorption of these intermediates. The Cu-Cu distances involved in $^*\text{CHCHO}^*$ adsorption are 2.48 Å and 2.74 Å (Fig. 4h), while those for $^*\text{CHC}^*\text{OH}$ adsorption are 2.43 Å and 3.26 Å (Fig. 4i). However, $^*\text{CHCHO}^*$ associated with Path 2 exhibits a thermodynamic preference over $^*\text{CHC}^*\text{OH}$ on Path 1 (-0.22 eV). For the Vs-CuS catalyst, two Cu sites are involved in the adsorption of intermediates, with the third Cu located at a distance of 3.85 Å. The Cu-Cu distance for $^*\text{CHCHO}^*$ is 3.58 Å (Fig. 4j and Supplementary Fig. 80a), whereas for $^*\text{CHC}^*\text{OH}$ adsorption it is 3.56 Å (Fig. 4k and Supplementary Fig. 80b). Moreover, $^*\text{CHC}^*\text{OH}$ associated with Path 1 exhibits a thermodynamic advantage over $^*\text{CHCHO}^*$ on Path 2 (-0.22 eV).

The site-projected d -PDOS was computed and ε_d was obtained as the first moment of the d -states referenced to E_F within a unified energy window. The results are summarized in Supplementary Fig. 81 (Co-sub-CuS) and Supplementary Fig. 82 (Vs-CuS). To capture the site that predominantly governs adsorption on each surface, we report the minimum ε_d among the adsorbing Cu sites ($\varepsilon_{d,\text{min}}$) as a representative descriptor for that adsorbate-surface pair. For Co-sub-CuS, the ε_d at the Cu_3 site reaches the lowest value ($\varepsilon_{d,\text{min}} = -2.711$ eV) for adsorption of CH-CH-O , and this coincides with the stronger adsorption of CH-CH-O on Co-sub-CuS relative to CH-C-OH (Supplementary Fig. 81). For Vs-CuS, the ε_d at the Cu_2 site is the lowest ($\varepsilon_{d,\text{min}} = -2.553$ eV) for adsorption of CH-C-OH , consistent with more favorable adsorption of CH-C-OH on Vs-CuS compared with CH-CH-O (Supplementary Fig. 82). Taken together, the Co-doping induced redistribution of ε_d across surface Cu sites switches the adsorption preference from CH-C-OH

(Vs-CuS) to CH-CH-O (Co-sub-CuS). These additions enhance the analysis and provide a site-level electronic rationale for the observed selectivity. Additionally, differential charge density analysis reveals that electron density accumulation between Cu sites and the adsorbed states of $^*\text{CHCHO}^*$ is significantly stronger than that of $^*\text{CHC}^*\text{OH}$, indicating a more robust covalent bond at the Cu sites and $^*\text{CHCHO}^*$ interface (Fig. 4l, Supplementary Fig. 83 and Supplementary data 1). Therefore, the Co-Sub-CuS catalyst exhibited the selectivity towards EtOH production via Path 2 compared to the Vs-Cu catalyst.

Discussion

In summary, Co subsurface-doped CuS with rich sulfur vacancies on the surface was synthesized via a one-pot redox solvothermal method. Combining in-situ isotopic spectroscopy and DFT calculation prediction, it was found that subsurface Co doping effectively regulates dual sites at an optimal distance, creating asymmetric active sites through surface Vs and surface-subsurface electron interactions. The asymmetric coupling of OCCOH was triggered by promoting a mixed adsorption configuration of $^*\text{CO}_{\text{atop}}$ and $^*\text{CO}_{\text{bridge}}$. Particularly, the electrons transfer from Cu to Co induced by subsurface Co doping enhances the oxophilicity of surface metal sites, which in turn facilitates the formation and conversion of $^*\text{CHCHO}^*$ intermediates in the form of surface-O bonds. This mechanism directs the subsequent protonation towards the EtOH pathway, and capturing the transition from $^*\text{CHCO}$ to $^*\text{CHCHO}^*$. As expected, the optimal $\text{Co}_{0.050}\text{-Sub-CuS}$ catalysts exhibited a FE_{EtOH} of nearly 80.0% with a partial current density of 602.0 mA cm^{-2} in a flow cell.

In an MEA electrolyzer, the FE_{EtOH} was approximately 78.7% consistently reproduced, with a $EtOH\ EE_{full-cell}$ value of 26.1% at -3.47 V, and stable electrolysis performance sustained for 305 h at $700\ mA\ cm^{-2}$. This work provides an in-depth understanding of the directional selectivity of $EtOH$ in CO_2ER , and may inspire the rational design of catalysts for the controlled CO_2ER to other high-value products.

Methods

Chemicals

Copper chloride dihydrate ($CuCl_2 \cdot 2H_2O$, 99.9%), thiourea (CH_4N_2S , 99.5%), cobalt(III) fluoride (CoF_3), cobalt(II) fluoride (CoF_2), iridium oxide (IrO_2 , 99.9%), poly(vinylidene fluoride) (PVDF), deuterium oxide (D_2O , 99.9%), ethanol (C_2H_5OH) and potassium hydroxide (KOH) were purchased from Aladdin Co. China. All reagents were used without further purification.

Synthesis of Co_x -Sub-CuS, $Co_{0.050}$ -Sur-CuS and CuS Nanosheets

First, dissolve $CuCl_2 \cdot H_2O$ (2 mmol) and Tu (4 mmol) separately in 20 mL of ethanol and sonication 10 min for dispersed, respectively. The two fresh solutions were mixed, and add the desired amount of CoF_3 (serving as cobalt-dopant) to the mixed solution under stirred. Then, the mixed solution was stirred for 15 min to form the white flocculent Co-doped $[Cu(Tu)_2]Cl$ precursor, which will serve as self-sacrificial templates for further vulcanization processes. Afterwards, transfer the flocculent solution into a 100 mL Teflon-lined stainless-steel autoclave and heat at $140\ ^\circ C$ for 15 h. After the reaction, the autoclave was naturally cooled to room temperature ($25^\circ C$) and centrifuge the resultant dark green product, washed with deionized water and ethanol for several times, respectively. Finally, the collected dark green products were dried in a vacuum oven for 12 h to obtain Co_x -sub-CuS nanosheets. In order to obtain Co_x -sub-CuS with different doping ratios, the mass of cobalt-dopant is controlled to 3.0, 6.0, 10.0, 15.0 and 30.0 mg to obtain $Co_{0.025}$ -Sub-CuS, $Co_{0.050}$ -Sub-CuS, $Co_{0.075}$ -Sub-CuS, and $Co_{0.100}$ -Sub-CuS samples, respectively.

For the synthesis of $Co_{0.050}$ -Sur-CuS sample, follow the same reaction conditions as the Co_x -sub-CuS synthesis, replacing 0.010 mmol of CoF_3 dopant with CoF_2 .

For the synthesis of CuS sample, $CuCl_2 \cdot H_2O$ (2 mmol) and Tu (4 mmol) separately in 20 mL of ethanol and sonication 10 min for dispersed, respectively. The two fresh solutions were mixed and stirred for 15 min to form the white flocculent $[Cu(Tu)_2]Cl$ precursor. Afterwards, transfer the flocculent solution into a 100 mL Teflon-lined stainless-steel autoclave and heat at $140\ ^\circ C$ for 15 h. After reaction, the autoclave was naturally cooled to room temperature ($25^\circ C$) and the resulting

dark green product was centrifuged, washed with deionized water and ethanol for several times and dried in a vacuum oven for 12 h to obtain pristine CuS sample.

Characterizations

X-ray diffraction (XRD) patterns were recorded using a SmartLab 9KW vertical powder X-ray diffractometer with a Cu-K α radiation. Scanning electron microscopy (SEM, Hitachi 8100), energy-dispersive spectroscopy (EDS, Hitachi 8100), and transmission electron microscopy (TEM, FEI Tecnai G220) were used to observe the sample morphology. Elemental contents in catalysts and electrolytes were determined using ICP-OES (iCAP PRO X). Chemical component analysis of synthesized catalysts was performed using X-ray photoelectron spectrometer (XPS) (Escalab 250Xi, Thermo Fisher). High-resolution TEM (HRTEM) and EDS mapping images were collected on a Talos F200S-G2 field-emission high-resolution transmission electron microscope operated at 200 kV. XANES and EXAFS experiments on Cu and Co K-edge were conducted at beamline BL14W1 in the Shanghai Synchrotron Radiation Facility (SSRF), China. TOF-SIMS experiments were conducted on a PHI nano TOF II, 5 iontof. The surface hydrophobicity of gas diffusion electrode (GDE) loaded with the catalyst were measured by Contact angle goniometer (CA, Attension Theta). EPR data were collected using a Bruker EMX PLUS.

Electrochemical Measurements

A three-chamber flow cell consisting of a gas chamber, catholyte chamber, and anolyte chamber was used for all electrochemical analyses (1 cm^2), the volume of each compartment was 1 cm^3 . The ambient temperature for electrochemical testing was maintained at $25\text{ }^\circ\text{C}$. The gas chamber and catholyte chamber were separated by the cathode electrode, while the catholyte and anolyte chambers were separated by a $1\times 1\text{ cm}^2$ anion-exchange membrane (FAB-PK-130, Fumasep Co., China) to enable product separation and ionic conduction. Prior to use, the membrane was first rinsed with deionized water until neutrality was achieved, then immersed in 1 M NaOH solution for 12 h for activation. Finally, the membrane was rinsed again with deionized water to remove residual electrolyte, and stored in deionized water for later use. Immediately before use, surface moisture was gently blotted with filter paper. A catalytically deposited gas diffusion electrode (GDE) and Hg/HgO were used as working and reference electrodes, respectively. The Hg/HgO electrode was calibrated against a stable standard saturated calomel electrode (0.241 V at $25\text{ }^\circ\text{C}$). A 1 M KOH electrolyte was prepared, and Ar was purged through it for 30 min to remove dissolved gases (e.g., O_2 and CO_2). The potential of the Hg/HgO electrode relative to the standard electrode was measured, and determined its standard electrode potential. To fabricate the working electrode, disperse 5 mg of catalysts in $250\text{ }\mu\text{L}$ ethanol with $15\text{ }\mu\text{L}$ of 5 wt% Nafion solution and ultrasonicated for 30 min to produce a mixed ink. Drop this mixture onto a gas diffusion layer (SGL carbon) with a geometric size of $1\times 1\text{ cm}^2$ to obtain a GDE with a loading amount controlled to $\sim 1.46\text{ mg cm}^2$ (The loading capacity was estimated based on the mass change of the gas diffusion layer before and after loading). Iridium oxide deposited Ti felt was used as the anode to avoid corrosion at oxidation potential (the preparation consistent with the working electrode). Place the working electrode between the gas and catholyte chambers to ensure gaseous

CO₂ diffusion and reaction at the catholyte/catalyst interface. Insert the reference electrodes into the catholyte chamber and maintain a specified distance from the working electrode. A freshly prepared 1 M KOH solution was used as the electrolyte, with a pH of approximately 13.8–14. An electrochemical workstation (CHI 760E, Shanghai Chenhua, China) with a current amplifier (CHI 680D, Shanghai Chenhua, China) was used for the CO₂ER test. Circulate 1 M KOH solution around the anode and cathode at a flow rate of 20 mL min⁻¹ via a peristaltic pump during CO₂ER. Supply CO₂ to the gas chambers at a constant flow rate of 30 mL min⁻¹ via a mass-flow controller. Collect LSV polarization curves at a scanning rate of 5 mV s⁻¹ in Ar and CO₂ atmospheres. Assess the long-term stability test in a flow cell at -0.8 V for more than 410 h with the environmental temperature maintained at 25 °C. In order to avoid the effect of carbonate precipitation on the product exchange in the anion exchange membrane, the electrolyte in the anode and cathode pools is changed every 24 h. The anion exchange membrane is replaced every 96 h to avoid the effect of membrane consumption on stability at high current density. All tested potentials were converted by the following formula (1):

$$E(\text{RHE}) = E(\text{Hg/HgO}) + 0.059 \times \text{pH} + E_0(\text{Hg/HgO}) + 0.85 iR \quad (1)$$

where $E(\text{Hg/HgO})$ is the potential against Hg/HgO measured in the experiment, and $E_0(\text{Hg/HgO})$ is the standard potential of Hg/HgO at 25°C (0.098 V). The ohmic loss between the working and reference electrodes was measured using electrochemical impedance spectroscopy (EIS) (frequency range: 100 kHz to 0.1 Hz, amplitude: 10 mV), and 85% iR compensation was applied to correct the potentials manually.

CO₂ER Test in an MEA Electrolyzer

Electroreduction of CO₂ in MEA consisted of two titanium backplates (TA2, Gaossunion Co., Ltd.) with a 1×1 cm² serpentine flow field. The cathode was prepared in the same way as the GDE in a flow cell (~1.46 mg cm⁻² for Co_{0.050}-Sub-CuS). Using iridium oxide (IrO₂) deposited Ti felt as the anode to avoid corrosion at oxidation potential—a material inert to ethanol oxidation yet highly active for the oxygen evolution reaction (OER)—as the anode catalyst can mitigate the cross-contamination and reoxidation of ethanol products. This, in turn, prevents the degradation of ethanol yield and MEA system performance. The cathode and anode were pressed onto the sides of the anion exchange membrane (Sustainion 37-50, Dioxide Materials), minimizing the gap between the electrodes to reduce ohmic losses. Gaseous CO₂ was fed through the gas diffusion layer at a constant flow rate (30 mL min⁻¹) via a mass flow controller, contact the catalyst. The 1 M KOH solution was used as the anolyte circulated via a gas-liquid mixed flow pump at a flow rate of 20 mL min⁻¹ during CO₂ER. By regulating the flow rate of the anode electrolyte, ethanol that permeates through the membrane can be promptly flushed away from the anode catalyst surface. This reduces ethanol's residence time at the anode and minimizes its oxidation probability. To enhance the solubility of inorganic salts and prevent salting-out, the reaction temperature was maintained at 60 °C throughout the experiment. The CO₂ER performance for MEA was evaluated by applying different currents using a 680D current amplifier in a two-electrode system on the CHI 760E electrochemical workstation. Long-term stability testing in a MEA electrolyzer was evaluated

at 700 mA cm⁻² for over 305 h. To avoid the effect of carbonate precipitation on the product exchange in the anion exchange membrane, the electrolyte in the anode and cathode pools is changed every 24 h, which ensures that ethanol which has diffused into the anode is recovered as much as possible before it undergoes complete oxidation. The anion exchange membrane is replaced every 48 h to avoid the effect of membrane consumption on stability at high current density. Cathodic gas products were vented through a simplified cold-trap to collect permeable liquid prior to gas chromatograph testing. FE values for the liquid products were computed based on the total mass of product collected on the anode and cathode.

CO₂ER product analysis

The gas products from CO₂ reduction (H₂, CO, CH₄, C₂H₄) were analyzed using a gas chromatograph (GC-9790II, Fuli Co., China) coupled with a thermal conductivity detector (TCD) and a flame ionization detector (FID). High-purity Ar (99.999%) was used as the carrier gas. The GC was calibrated based on the certified standard gas samples including H₂, CO, and C₁ to C₃ hydrocarbons (Xuancheng Liyuan Gas Co., Ltd). The liquid products were analyzed by ¹H nuclear magnetic resonance (NMR) spectrometer (Bruker ASCEnd TM 400MHz), in which 500 μL of catholyte after CO₂ER was collected and mixed with 100 μL of DMSO (internal standard, diluted to 10000 ppm (v/v) by deuterated water), and the ZGPR method was used to suppress the water peak. All experimental results were repeated at least three times, keeping all conditions constant.

The partial current densities for gas products were calculated by the following formula (2):

$$j_i = (n_i \times v \times z_i \times F \times p_0) / (R \times T) \times (A)^{-1} \quad (2)$$

The Faradaic efficiency for gas products was calculated by the following formula (3):

$$FE_i = j_i / j_{\text{total}} \times 100\% \quad (3)$$

where n_i is the number of electrons required to be transferred to generate the corresponding product i , v is the gas flow rate measured by a mass-flow controller (S49 33/MT) at the exit of the electrochemical cell (mL min⁻¹), z_i is the volume fraction of the gas product based on a calibration of the online gas chromatography, F is the Faradaic constant (96485 C mol⁻¹), p_0 is the standard atmospheric pressure (101.3 kPa), R is the gas constant (8.314 J mol⁻¹ K⁻¹), T is the temperature (298.15 K) and A is the geometric area of the working electrode (cm²)

The Faradaic efficiency for liquid products was calculated by the following formula (4):

$$FE_i = (c_i \times V \times n_i \times F) / (I \times T) \times 100\% \quad (4)$$

The partial current densities for liquid products were calculated by the following formula (5):

$$j_i = FE_i \times j_{\text{total}} \quad (5)$$

where c_i is the concentration of i product, V is the electrolyte volume, I is the total current density, T is total reaction time.

Half-cell energy efficiency

Half-cell energy efficiency ($EE_{1/2}$) was computed on basis of the cathodic CO₂ER coupled with the anodic water oxidation reaction ($\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \leftrightarrow 2\text{H}_2\text{O}$; 1.23 V vs. RHE) (6):

$$EE_{1/2} = (E_{\text{oxo}} - E_{\text{redo}}) / (E_{\text{ox}} - E_{\text{red}}) \times FE \quad (6)$$

where E_{oxo} and E_{redo} are the thermodynamic potentials for water oxidation and CO₂ER to products,

respectively. E_{ox} and E_{red} are the applied potentials at anode and cathode, respectively. For computation of the half-cell EE , the anodic reaction was assumed to occur with an overpotential of 0 V, that means $E_{ox} = 1.23$ V.

Full-cell energy efficiency

The full-cell energy efficiency for product i is calculated as follows (7):

$$EE_{full-cell} = ((1.23 + (-E_0)) \times FE_i) / (-E_{full-cell}) \times FE_i \quad (7)$$

where E_0 is the thermodynamic potential of CO_2 to product A calculated based on the standard molar Gibbs energy formation at 298.15 K, FE_i is the measured FE (%) of the product i , and $E_{full-cell}$ is the full-cell voltage measured in the MEA system without ohmic loss correction.

ECSA measurement

ECSA was measured by double-layer capacitance (C_{dl}) methods. All experiments are conducted in the flow cell and the used catalysts are obtained following activation under CO_2 ER. C_{dl} and obtained from CV curves at the potential range of 0.1-0.2 V vs Ag/AgCl with the stepwise rates from 10 to 50 mV s⁻¹. The capacitance currents at 0.15 V vs RHE were plotted against the scan rates, and C_{dl} (mF cm⁻²) was derived from the slope according to the following (8):

$$C_{dl} = \Delta J / v \quad (8)$$

where ΔJ is the capacitance current (half of the difference between the anodic current density and cathodic current density, $(J_a - J_c)/2$), and v is the scan rate.

ECASAs were calculated by the equation (9):

$$ECSA = C_{dl} / C_s \times A \quad (9)$$

where C_s is the capacitance of a smooth planar surface (0.04 mF cm⁻² in alkaline media), and A is the electrode area.

Electrochemical in-situ FTIR spectra measurements

In-situ FTIR spectra tests were conducted in a spectroelectrochemical cell with a three-electrode configuration. The cell was integrated into a Bruker INVENIO R Fourier-transform infrared spectrometer equipped with a liquid nitrogen-cooled MCT detector. The gas diffusion electrode loaded with the catalyst was used as working electrode. A Pt wire and Ag/AgCl electrodes were used as the counter and reference electrodes, respectively. The in-situ isotope spectra tests were conducted under the following conditions: ¹²CO₂-saturated 1 M KOH with H₂O, ¹³CO₂-saturated 1 M KOH with H₂O, and ¹²CO₂-saturated 1 M KOH with D₂O. To prepare the 1 M KOH electrolyte, KOH was dissolved in either H₂O or D₂O. Prior to the tests, the system was purged with the corresponding isotope-labeled CO₂ gas for ten min to eliminate interference from other gases. When testing under the ¹²CO₂-saturated 1 M KOH with D₂O conditions, both the working electrode and the entire cell were thoroughly rinsed several times with D₂O to remove residual H₂O, followed by vacuum drying at 60 °C. A fixed-angle Ge prism (60°) coated with catalysts embed into the bottom of the cell. Before testing, the detector was cooled with liquid nitrogen for at least 30 min to maintain a stable signal. Prior to measurements, the blank spectrum was tested under OCP conditions for baseline correction, and then the spectrums under different

applied potentials were collected at a resolution of 4 cm^{-1} . The Spectrums were presented in absorbance, in which the positive and negative peaks show the increase and decrease in signals, respectively. All spectroscopic data were baseline corrected and subtracted with the background.

Electrochemical in-situ Raman spectra tests

In-situ Raman experiments were performed using a Raman Spectrophotometer RFS 100/S (Bruker Co., Germany) in a homemade standard three-electrode system with a quartz window by GaossUnion Photoelectric, China, and a Nd/YAG laser of the 532 nm line was used as the excitation source. A Pt wire and Ag/AgCl electrodes were used as the counter and reference electrodes, respectively. The gas diffusion electrode loaded with the catalyst was used as working electrode. The 1 M KOH solution was used as electrolyte. Supply CO_2 into gas chambers at a constant flow rate of 30 mL min^{-1} via a mass-flow controller. Then, the Raman signals were recorded after 15 min electrolysis at OCP and different applied potentials, all spectroscopic data were baseline corrected and subtracted with the background.

Computational methods

Density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP) with the projector augment wave (PAW) method⁵⁶. The electron exchange and correlation energy were treated within the generalized gradient approximation (GGA) and the Perdew, Bueke, and Ernzerhof (PBE) functional. Spin polarization was taken into account in all calculations. The Brillouin zone was sampled using a k-points grid of $1\times 1\times 1$. The energy cut-off and convergence criteria for force were set as 480 eV and 0.03 eV/Å, respectively. Partial occupancies were determined using a Gaussian smearing scheme with a smearing width of 0.05 eV. The ground state structures of the adsorbed $^*\text{COCO}$, $^*\text{CHCO}$, $^*\text{CHCOH}$, $^*\text{CHCHO}$, $^*\text{CHC}$ and $^*\text{OCHCH}_2$ were determined by searching the lowest energy one among all the possible configurations. The CuS (100) surface was modeled by six atomic layers of 72 atoms, and the geometry of the upper three layers was optimized while constraining the lower three layers. The thickness of the vacuum layer is 20 Å, and the supercell dimensions are 11.447 Å, 16.463 Å and 25.507 Å. For the Vs- CuS model, a sulfur atom in the first atomic layer of CuS (100) surface was removed. For the Co-sub-CuS model, a copper atom in the second atomic layer of CuS-Vs surface was replaced with a cobalt atom. Density of states were calculated and weighted by the respective Hamilton matrix elements using the Local-Orbital Basis Suite Towards Electronic-Structure Reconstruction software package (LOBSTER)⁵⁷. The anti-bonding and bonding interactions were analyzed based on the projected crystal orbital Hamilton population (-pCOHP) method. Changes in reaction energy for the competing pathways leading to C_2 products were calculated.

We calculated the activation barriers for asymmetric coupling of $^*\text{CO}$ and $^*\text{COH}$ to form $^*\text{OCCOH}$ using the dimer method implemented in the CP2K code with the Quickstep module⁵⁸. The molecularly optimized Double-Zeta-Valence-Polarization (DZVP-MOLOPT) basis sets and Goedecker-Teter-Hutter (GTH) pseudopotentials were used for core electrons in combination with Perdew-Burke-Ernzerhof (PBE) exchange correlation functional⁵⁹. All calculations were converged with a Broyden-Fletcher-Goldfarb-Shannon (BFGS) optimizer. The periodic boundary

conditions were used with a plane-wave grid cutoff of 400 Ry and the Grimme-D3 dispersion correction. Vibrational frequency calculation was performed to ascertain the saddle state by confirming the existence of an imaginary vibrational frequency. To compute vibrational spectra, normal-mode analysis was performed by evaluating second derivatives of energy concerning nuclear displacements using MOLDEN package⁶⁰. AIMD simulations were performed on the optimized structures with the CP2K package in the constant-volume and constant-temperature (NVT) ensemble. The temperature was controlled with Nosé-Hoover thermostat at 300 K. The AIMD simulations are run for 10 ps with the time step of 1 fs. The radial distribution function (RDF) analysis was performed using VMD program.

Data availability

All data that support the findings in this paper are available within the article and its Supplementary Information or are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

References

1. A. Ozden *et al.*, Energy- and carbon-efficient CO₂/CO electrolysis to multicarbon products via asymmetric ion migration–adsorption. *Nat. Energy* **8**, 179-190 (2023).
2. S. Dey, F. Masero, E. Brack, M. Fontecave, V. Mougél, Electrocatalytic metal hydride generation using CPET mediators. *Nature* **607**, 499-506 (2022).
3. P. De Luna *et al.*, What would it take for renewably powered electrosynthesis to displace petrochemical processes? *Science* **364**, 350 (2019).
4. M. Zhong *et al.*, Accelerated discovery of CO₂ electrocatalysts using active machine learning. *Nature* **581**, 178-183 (2020).
5. Cao-Thang Dinh *et al.*, CO₂ electroreduction to ethylene via hydroxide-mediated copper catalysis at an abrupt interface. *Science* **360**, 783–787 (2018).
6. Y. Y. Birdja *et al.*, Advances and challenges in understanding the electrocatalytic conversion of carbon dioxide to fuels. *Nat. Energy* **4**, 732-745 (2019).
7. R. M. Arán-Ais, F. Scholten, S. Kunze, R. Rizo, B. Roldan Cuenya, The role of in situ generated morphological motifs and Cu(i) species in C₂₊ product selectivity during CO₂ pulsed electroreduction. *Nat. Energy* **5**, 317-325 (2020).
8. H. Zhang, J. Gao, D. Raciti, A. S. Hall, Promoting Cu-catalysed CO₂ electroreduction to multicarbon products by tuning the activity of H₂O. *Nat. Catal.* **6**, 807-817 (2023).

9. W. Ma *et al.*, Electrocatalytic reduction of CO₂ to ethylene and ethanol through hydrogen-assisted C–C coupling over fluorine-modified copper. *Nat. Catal.* **3**, 478-487 (2020).
10. S. Nitopi *et al.*, Progress and Perspectives of Electrochemical CO₂ Reduction on Copper in Aqueous Electrolyte. *Chem. Rev.* **119**, 7610-7672 (2019).
11. P. Wei *et al.*, Coverage-driven selectivity switch from ethylene to acetate in high-rate CO₂/CO electrolysis. *Nat. Nanotechnol.* **18**, 299-306 (2023).
12. J. M. Wang *et al.*, Asymmetric gradient orbital interaction of hetero-diatomic active sites for promoting C - C coupling. *Nat. Commun.* **14**, 3808 (2023).
13. A. Vasileff, C. Xu, Y. Jiao, Y. Zheng, S.-Z. Qiao, Surface and Interface Engineering in Copper-Based Bimetallic Materials for Selective CO₂ Electroreduction. *Chem* **4**, 1809-1831 (2018).
14. M. Chhetri *et al.*, Dual-site catalysts featuring platinum-group-metal atoms on copper shapes boost hydrocarbon formations in electrocatalytic CO₂ reduction. *Nat. Commun.* **14**, 3075 (2023).
15. J. Zhang *et al.*, Steering CO₂ electroreduction pathway toward ethanol via surface-bounded hydroxyl species-induced noncovalent interaction. *Proc. Natl. Acad. Sci. U. S. A.* **120**, e2218987120 (2023).
16. Y. Zheng *et al.*, Understanding the Roadmap for Electrochemical Reduction of CO₂ to Multi-Carbon Oxygenates and Hydrocarbons on Copper-Based Catalysts. *J. Am. Chem. Soc.* **141**, 7646-7659 (2019).
17. H. J. Peng, M. T. Tang, J. Halldin Stenlid, X. Liu, F. Abild-Pedersen, Trends in oxygenate/hydrocarbon selectivity for electrochemical CO₂ reduction to C₂ products. *Nat. Commun.* **13**, 1399 (2022).
18. T. Cheng, H. Xiao, W. A. Goddard, Full atomistic reaction mechanism with kinetics for CO reduction on Cu(100) from ab initio molecular dynamics free-energy calculations at 298 K. *Proc. Natl. Acad. Sci. U. S. A.* **114**, 1795-1800 (2017).
19. H. Peng *et al.*, The role of atomic carbon in directing electrochemical CO₂ reduction to multicarbon products. *Energy Environ. Sci.* **14**, 473-482 (2021).
20. S. Zhao *et al.*, Steering carbon dioxide reduction toward C-C coupling using copper electrodes modified with porous molecular films. *Nat. Commun.* **14**, 844 (2023).
21. W. Ge *et al.*, Dynamically Formed Surfactant Assembly at the Electrified Electrode-Electrolyte Interface Boosting CO₂ Electroreduction. *J. Am. Chem. Soc.* **144**, 6613-6622 (2022).
22. J. Feng *et al.*, Improving CO₂-to-C₂₊ Product Electroreduction Efficiency via Atomic Lanthanide Dopant-Induced Tensile-Strained CuO_x Catalysts. *J. Am. Chem. Soc.* **145**, 9857-9866 (2023).

23. Y. C. Li *et al.*, Binding Site Diversity Promotes CO₂ Electroreduction to Ethanol. *J. Am. Chem. Soc.* **141**, 8584-8591 (2019).
24. C. Long *et al.*, Regulating reconstruction of oxide-derived Cu for electrochemical CO₂ reduction toward n-propanol. *Sci. Adv.* **9**, eadi6119 (2023).
25. B. Jia *et al.*, Restraining Interfacial Cu²⁺ by using Amorphous SnO₂ as Sacrificial Protection Boosts CO₂ Electroreduction. *Adv. Mater.* **35**, e2305587 (2023).
26. W. H. Lai *et al.*, Activating Inert Surface Pt Single Atoms via Subsurface Doping for Oxygen Reduction Reaction. *Nano Lett.* **21**, 7970-7978 (2021).
27. W. Wei *et al.*, Subsurface oxygen defects electronically interacting with active sites on In₂O₃ for enhanced photothermocatalytic CO₂ reduction. *Nat. Commun.* **13**, 3199 (2022).
28. S. Wang *et al.*, Subsurface-Carbon-Induced Local Charge of Copper for an On-Surface Displacement Reaction. *Angew. Chem. Int. Ed.* **60**, 23123-23127 (2021).
29. K. Yao *et al.*, Mechanistic Insights into OC-COH Coupling in CO₂ Electroreduction on Fragmented Copper. *J. Am. Chem. Soc.* **144**, 14005-14011 (2022).
30. H. Zhao *et al.*, The role of Cu₁-O₃ species in single-atom Cu/ZrO₂ catalyst for CO₂ hydrogenation. *Nat. Catal.* **5**, 818-831 (2022).
31. Z. Ma *et al.*, CO₂ electroreduction to multicarbon products in strongly acidic electrolyte via synergistically modulating the local microenvironment. *Nat. Commun.* **13**, 7596 (2022).
32. W. He *et al.*, Enhanced Nitrate-to-Ammonia Efficiency over Linear Assemblies of Copper-Cobalt Nanophases Stabilized by Redox Polymers. *Adv. Mater.* **35**, e2303050 (2023).
33. J. Zhou *et al.*, Linking oxidative and reductive clusters to prepare crystalline porous catalysts for photocatalytic CO₂ reduction with H₂O. *Nat. Commun.* **13**, 4681 (2022).
34. S. F. Hung *et al.*, Unraveling Geometrical Site Confinement in Highly Efficient Iron-Doped Electrocatalysts toward Oxygen Evolution Reaction. *Adv. Energy Mater.* **8**, 1701686 (2017).
35. Z. Xiao *et al.*, Filling the oxygen vacancies in Co₃O₄ with phosphorus: an ultra-efficient electrocatalyst for overall water splitting. *Energy Environ. Sci.* **10**, 2563-2569 (2017).
36. Z. Liu *et al.*, Interfacial Water Tuning by Intermolecular Spacing for Stable CO₂ Electroreduction to C₂₊ Products. *Angew. Chem. Int. Ed.* **62**, e202309319 (2023).
37. M. Jouny, W. Luc, F. Jiao, High-rate electroreduction of carbon monoxide to multi-carbon products. *Nat. Catal.* **1**, 748-755 (2018).
38. P. Wang *et al.*, Boosting electrocatalytic CO₂-to-ethanol production via asymmetric C-C coupling. *Nat. Commun.* **13**, 3754 (2022).
39. C. Zhu *et al.*, Dynamic restructuring of epitaxial Au-Cu biphasic interface for tandem CO₂-

- to-C₂₊ alcohol conversion. *Chem* **8**, 3288-3301 (2022).
40. X. Wang *et al.*, Ag⁺-Doped InSe Nanosheets for Membrane Electrode Assembly Electrolyzer toward Large-Current Electroreduction of CO₂ to Ethanol. *Angew. Chem. Int. Ed.* **62**, e202313646 (2023).
 41. S. J. Shin *et al.*, A unifying mechanism for cation effect modulating C1 and C2 productions from CO₂ electroreduction. *Nat. Commun.* **13**, 5482 (2022).
 42. G. Zhang *et al.*, Selective CO₂ electroreduction to methanol via enhanced oxygen bonding. *Nat. Commun.* **13**, 7768 (2022).
 43. S. Zhu, B. Jiang, W. B. Cai, M. Shao, Direct Observation on Reaction Intermediates and the Role of Bicarbonate Anions in CO₂ Electrochemical Reduction Reaction on Cu Surfaces. *J. Am. Chem. Soc.* **139**, 15664-15667 (2017).
 44. R. Amirbeigi *et al.*, Atomic-scale surface restructuring of copper electrodes under CO₂ electroreduction conditions. *Nat. Catal.* **6**, 837-846 (2023).
 45. H. Kang *et al.*, Generation of oxide surface patches promoting H-spillover in Ru/(TiO_x)MnO catalysts enables CO₂ reduction to CO. *Nat. Catal.* **6**, 1062-1072 (2023).
 46. B. Yang *et al.*, Electrocatalytic CO₂ reduction to alcohols by modulating the molecular geometry and Cu coordination in bicentric copper complexes. *Nat. Commun.* **13**, 5122 (2022).
 47. S. Kuang *et al.*, Asymmetrical electrohydrogenation of CO₂ to ethanol with copper-gold heterojunctions. *Proc. Natl. Acad. Sci. U. S. A.* **120**, e2214175120 (2023).
 48. E. Perez-Gallent, M. C. Figueiredo, F. Calle-Vallejo, M. T. Koper, Spectroscopic Observation of a Hydrogenated CO Dimer Intermediate During CO Reduction on Cu(100) Electrodes. *Angew. Chem. Int. Ed.* **56**, 3621-3624 (2017).
 49. Y. Shi *et al.*, Localized Geometry Determined Selectivity of Iodide-Derived Copper for Electrochemical CO₂ Reduction. *Adv. Energy Mater.* **13**, 2203896 (2023).
 50. D. L. Meng *et al.*, Highly Selective Tandem Electroreduction of CO₂ to Ethylene over Atomically Isolated Nickel-Nitrogen Site/Copper Nanoparticle Catalysts. *Angew. Chem. Int. Ed.* **60**, 25485-25492 (2021).
 51. L. Zhang *et al.*, Oxophilicity-Controlled CO₂ Electroreduction to C₂₊ Alcohols over Lewis Acid Metal-Doped Cu^{δ+} Catalysts. *J. Am. Chem. Soc.* **145**, 21945-21954 (2023).
 52. W. Sun *et al.*, V-Doped Cu₂Se Hierarchical Nanotubes Enabling Flow-Cell CO₂ Electroreduction to Ethanol with High Efficiency and Selectivity. *Adv. Mater.* **34**, e2207691 (2022).
 53. J. Ding *et al.*, A tin-based tandem electrocatalyst for CO₂ reduction to ethanol with 80% selectivity. *Nat. Energy* **8**, 1386-1394 (2023).

54. Wang, P. et al. Boosting electrocatalytic CO₂-to-ethanol production via asymmetric C–C coupling. *Nat. Commun.* **13**, 3754 (2022).
55. Kuang, S. et al. Asymmetrical electrohydrogenation of CO₂ to ethanol with copper–gold heterojunctions. *Proc. Natl. Acad. Sci. U. S. A.* **120**, e2214175120 (2023).
56. Kresse, G. & Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
57. Maintz, S., Deringer, V. L., Tchougréeff, A. L. & Dronskowski, R. LOBSTER: A tool to extract chemical bonding from plane-wave based DFT. *J. Comput. Chem.* **37**, 1030–1035 (2016).
58. Henkelman, G. & Jónsson, H. A dimer method for finding saddle points on high dimensional potential surfaces using only first derivatives. *J. Chem. Phys.* **111**, 7010–7022 (1999).
59. Krack, M. Pseudopotentials for H to Kr optimized for gradient-corrected exchange-correlation functionals. *Theor. Chem. Acc.* **114**, 145–152 (2005).
60. Schaftenaar, G. & Noordik, J. H. Molden: a pre- and post-processing program for molecular and electronic structures. *J. Comput. Aid. Mol. Des.* **14**, 123–134 (2000).

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Author Contributions

M.Z.G., G.F.W., X.J.Z., J.J.C. and H.Q.Y. conceived and designed the research; M.Z.G. M.Z.G. conducted the synthesis and characterization. M.Z.G. and F.Z. analyzed the data and relevant results. Y.M. performed the DFT calculations. M.Z.G. and L.J. performed the electrochemical experiments. M.Z.G. and Q.C. performed and discussed soft X-ray absorption spectroscopy characterization. G.F.W., X.J.Z., J.J.C. and H.Q.Y. contributed to the planning and coordination of the project. M.Z.G., F.Z., G.F.W., X.J.Z., J.J.C. and H.Q.Y. co-wrote and edited the manuscript. All the authors contributed to discussion of the results and the manuscript.

Competing interests

The authors declare no competing interests.

Additional Information

Supplementary information The online version contains supplementary material available at <https://www.nature.com/natcom>.

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Figure Captions

Fig. 1 | Structural characterizations. (a) TEM images of Co_{0.050}-Sub-CuS. (b) HRTEM images of Co_{0.050}-Sub-CuS. (b1-b3) enlargement of dash squares in (b), S-vacancies (V_S) (labeled by the yellow dotted circles) and line defect (marked by the dotted red line). (c-e) FFT patterns of respective region. Cobalt TOF-SIMS mapping images of Co_{0.050}-Sub-CuS (f) and Co_{0.050}-Sur-CuS (g), (insert, semiquantitative analysis of TOF-SIMS spectra of Co element of Co_{0.050}-Sub-CuS and Co_{0.050}-Sur-CuS). (h) Cu K-edge XANES spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS and CuS. (i) Fitting of k^2 -weighted EXAFS spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS and CuS. (j) Co K-edge XANES spectra of Co_{0.050}-Sub-CuS, Co_{0.050}-Sur-CuS and standards. (k) Fitting of k^2 -weighted EXAFS spectra of Co_{0.050}-Sur-CuS, Co_{0.050}-Sub-CuS and standards. (l) Corresponding wavelet transforms of k^2 -weighted EXAFS spectra at Co K-edge for Co_{0.050}-Sub-CuS and Co_{0.050}-Sur-CuS. (m) Scheme of activating and enhancing oxophilicity of surface Cu sites by subsurface Co doping.

Fig. 2 | CO₂ER performance for Co_x-Sub-CuS, Co_{0.050}-Sur-CuS and CuS. (a) LSV curves. (b) Potential-dependent C₂ FE and EtOH current density. (c) Ratio of FE_{C₂} to FE_{CO} and (d) Ratio of FE_{EtOH} to FE_{C₂H₄} under selected potential. FE (e), yield rate (f) and EE (g) for EtOH production over Co_x-Sub-CuS and Co_{0.050}-Sur-CuS. (h) Schematic diagram of the MEA electrolyzer. (i) C₂ FE and partial current density at selected cell voltage under MEA measurement. (j) Comparison of the performance metrics of the Co_{0.050}-Sub-CuS catalyst with recently reported catalysts (details in Table S8). (k) CO₂ER stability of the Co_{0.050}-Sub-CuS catalyst in a three-electrode flow cell at a constant potential of 0.8 V vs. RHE. (l) EtOH FE and cell voltage during CO₂ER at a constant current of 700 mA cm⁻² in an MEA electrolyzer. All voltages were converted to the reversible hydrogen electrode (RHE) scale with 85% iR-correction. Error bars represent the standard deviation from at least three independent measurements.

Fig. 3 | In-situ characterization and mechanism exploration of the CO₂ER to EtOH. In-situ FTIR spectra of Co_{0.050}-Sub-CuS in (a) ¹²CO₂-saturated 1 M KOH with H₂O, (b) ¹³CO₂-saturated 1 M KOH with H₂O. (c) Comparison of the in-situ FTIR spectra on Co_{0.050}-Sub-CuS in ¹²CO₂-saturated 1 M KOH with H₂O, ¹³CO₂-saturated 1 M KOH with H₂O and ¹²CO₂-saturated 1 M KOH with D₂O. In-situ FTIR spectra of Co_{0.050}-Sub-CuS in (d) ¹²CO₂-saturated 1 M KOH with D₂O as electrolyte from 0.1 V vs. RHE to -1.1 V vs. RHE. (e) The calculated infrared-active vibrational frequency of *OCCOH on Co_{0.050}-Sub-CuS. (f) The calculated infrared-active vibrational frequency of *CHCHO* on Co_{0.050}-Sub-CuS, the right is the different adsorption modes of two key intermediates *CHCHO* and *CHC*OH at dual-Cu sites. (g) In-situ Raman peaks on Co_{0.050}-Sub-CuS in 1 M KOH from 0.1 V vs. RHE to -1.1 V vs. RHE during CO₂ER. (h) Comparison of the LFB/(HFB+LFB) ratio in in-situ Raman spectra for different catalysts at different applied potential. (i) Comparison of the *OC_{bridge}/(^{*}OC_{bridge}+^{*}OC_{atop}) ratio in in-situ FTIR spectra for different catalysts at different applied potential. (j) Schematic illustration of the underexplored asymmetric

coupling process and ethanol path directed selectivity problem in CO₂ electroreduction, and flow diagram illustrating the working principles of isotopic tracking of surface-adsorbed intermediates based on ¹²C/¹³C and H/D isotopes in tandem with in-situ FTIR spectra. (k) Schematic for boosted EtOH generation over Co_{0.050}-Sub-CuS. Blue, silver, pink, red, gray, and green-color spheres in the model represent Cu, S, Co, O, C, and H atoms, respectively.

Fig. 4 | Mechanistic study. (a) Reaction routes via C-C coupling (grey) towards C₂ products. Path 1 for the generation of ethylene (blue), and Path 2 for EtOH (red). (b) On the Co-Sub-CuS catalyst, *CO and *COH co-adsorb on Cu sites and asymmetrically couple to form OCCOH*, with dashed arrows denoting Cu-Cu distances. (c) Energy profiles for the *CO + *COH → OCCOH* on Vs-CuS and Co-Sub-CuS. Initial state (IS), transition state (TS) and final state (FS) are depicted in the insets. (d) Energy profiles for the reduction of *CHCO. (e) -pCOHP curves of the C-Cu₁ and C-Cu₂ bonds. (f) Heat map depicting reaction energy profiles for conversion of *CHCO. B₁-B₃ refer to three adsorption configurations of *CHCO; C₁-C₄ denote four configurations of *CHC*OH along Path 1 and O₁-O₄ represent *CHCHO* along Path 2. See Supplementary Information for structure details. (g) *CHCHO* on Co-Sub-CuS with one layer of water molecules. Hydrogen bonds are shown in dashed lines, with arrows illustrating the Cu-Cu distance in (h) *CHCHO* and (i) *CHC*OH. Configurations of (j) *CHCHO* and (k) *CHC*OH on Vs-CuS. (l) Differential charge density plots of *CHCHO* on Co-Sub-CuS. Isosurfaces were generated at ±0.002 e⁻/bohr³, with blue and yellow areas representing electron loss and gain, respectively.

Editorial Summary:

Precisely controlling the configuration of oxygen-binding intermediates is crucial for selective CO₂-to-ethanol electroreduction. Here, the authors present a subsurface Co-doped CuS catalyst, which transforms key intermediates via surface-O bonds to achieve directional selectivity for ethanol.

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